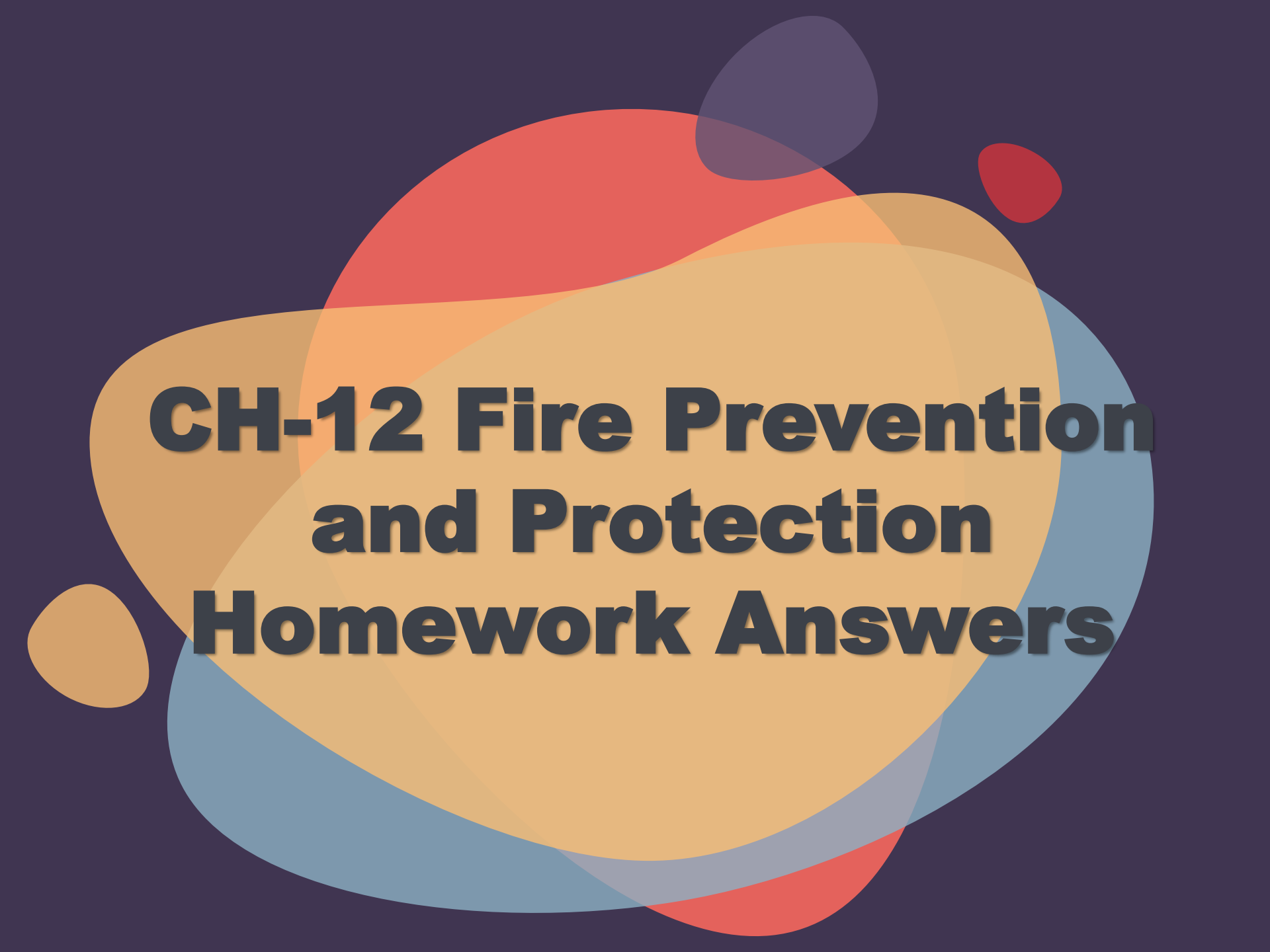




CH-12 Fire Prevention and Protection Homework Answers



01

The chemistry of fire is depicted by a fire triangle. Which of the following does NOT apply?

- A. Air, heat and fuel comprise the three elements of the triangle
- B. All three elements must be present to sustain combustion
- C. The combustion process will cease when any one of the three elements is removed
- D. **Water must be used to extinguish any fire**



02

In the theory of fire extinguishment, all of the following are believed appropriate, EXCEPT:

- A. Extinguishment by cooling
- B. Extinguishment by oxygen dilution
- C. Extinguishment by fuel removal
- D. Extinguishment by fire break**



03

The volatility of a chemical refers to the:

A. Capacity of a liquid to form vapors

B. Degree of flammability of a chemical

C. Exothermic reaction of a liquid

D. Endothermic reaction of a liquid



04

A flammable liquid is best defined by one of the following:

- A. Any liquid having a flash point below 73°F and a boiling point below 100°F
- B. Any liquid having a flash point below 100°F and a vapor pressure not exceeding 40 psi(a) at 100°F.**
- C. Any liquid having flash points at or above 100°F and below 140°F
- D. Any liquid having a flash point below 73°F and having a boiling point at or above 100°F



05

Vapors of flammable and combustible liquid are:

A. Generally heavier than air

B. Lighter than air

C. Normally liquid at room temperature and have molecular weights lower than those which are gases

D. Unstable and unpredictable



06

A flammable liquid with a flash point of 72°F and a boiling point below 100°F would be classified by the NFPA as a:

- A. Class I liquid
- B. Class IA liquid**
- C. Class IB liquid
- D. Class II liquid



07

The minimum concentration of gasoline in air that can be ignited is 7.6%. This minimum concentration is called the:

A. Lower Explosive Limit

B. Upper Explosive Limit

C. Lower ignitable limit

D. A and C

Flammable limits and range is defined as:

- A. The range in which flammable liquids cannot ignite
- B. The range in which flammable liquids can explode
- C. The range in which liquids form flammable mixtures with maximum concentrations of vapor in air or oxygen
- D. The range in which gases or vapors form flammable mixtures with air or oxygen**



09

Water principally extinguishes fires by:

A. Cooling the fuel surface

B. Excluding oxygen

C. Pre-wetting the fuel

D. Dilution



10

What stipulation must be met for the storage of flammable or combustible liquids in indoor areas?

- A. A clear pathway of at least 20 ft in width must be maintained.
- B. The air within the room must be completely replaced at a rate of six times per hour.**
- C. The storage containers must cover a minimum area of 600 ft²
- D. Storage containers with more than 30 gallons may be stacked